Ethical Simulation in Autonomous Driving

Abstract

This study tackles the ethical challenges faced by autonomous vehicles, particularly in life or death decisions following a fatality. The central question is whether AI can make morally aligned decisions similar to human preferences, especially in scenarios like the "trolley problem." A rule based ethical simulator was developed to model these decisions, incorporating factors such as ethical value, age, and AI preferences for younger or older individuals. A survey of 150 participants compared human choices with AI decisions, revealing significant alignment between the two. These findings suggest that AI can be programmed to make decisions consistent with human ethical inclinations, advancing the development of ethically responsible autonomous driving systems.

Introduction

This study investigates the ethical challenges associated with autonomous vehicles, focusing on the question of ethical and legal responsibility when an autonomous vehicle causes a fatality. While the legal aspects are inherently complex and beyond the scope of this research, the ethical dimension is explored in depth. The research aims to develop an ethical simulator for an Artificial Intelligence (AI) system deployed in autonomous vehicles, with a specific emphasis on the renowned ethical dilemma known as the "trolley problem." The simulation analyzes the decision making processes of the AI based on three key variables: ethical value, the age of the individuals involved, and the AI's preference between younger and older individuals. To achieve this, a rule based algorithm was designed to mimic decision making processes, functioning as a pseudo AI without preconceived biases. Furthermore, a parallel study was conducted, involving a survey of 150 participants to understand human decision making in comparable scenarios. The outcomes of the algorithm were then systematically compared with the human responses, providing valuable insights into the alignment between machine generated and human ethical decisions.

Methodology

We developed a C++ program to represent the AI of an autonomous vehicle, designed to model the trolley problem by simulating scenarios involving two groups of individuals, referred to as Group A and Group B. The AI is tasked with deciding which group to sacrifice, leveraging evaluation algorithms that incorporate two primary variables: the ethical weight assigned to the decision and the ages of the individuals involved. To ensure ethical neutrality, the AI evaluates both options without any inherent biases, enabling objective comparisons in each scenario.

This AI functions as a rule based ethical decision making algorithm, computing scores based on the combination of ethical weight and the age distribution within each group. This approach facilitates an unbiased decision making process, allowing the algorithm to systematically select between the groups in a manner that adheres to predefined ethical principles. The iterative simulation design enables the exploration of AI behavior in a range of complex moral situations, providing insights into how such systems could operate under real world constraints.

In parallel, a survey was conducted to gather data on human decision making preferences in similar scenarios. The survey involved a sample of 150 participants who were presented with a series of hypothetical dilemmas mimicking the trolley problem. The scenarios varied in key parameters, such as the number of individuals in each group, their ages, and the potential consequences of the AI's decisions. Participants were asked to indicate their preferred choice in each situation and provide reasoning for their decisions when applicable.

The survey results were analyzed to identify trends in human ethical reasoning, such as preferences for prioritizing certain age groups or minimizing total casualties. These findings were then juxtaposed with the AI's simulation outcomes to evaluate the alignment between human and algorithmic decision making.

Study Variables

The simulation incorporated a detailed set of variables to thoroughly analyze the ethical decision making processes of the AI. These variables were chosen to represent both the quantitative and qualitative aspects of moral dilemmas, providing a nuanced understanding of how the AI responds to complex scenarios.

One of the core variables is the ethical value assigned to outcomes, which serves as a measure of the moral weight attributed to each decision. This value, randomly assigned on a predefined scale from 1 to 5, influences how significant a particular choice is from an ethical perspective. Additionally, the ages of individuals within the groups, ranging between 5 and 80 years, play a critical role, with the AI adjusting its prioritization based on age specific weighting. The AI's decision making process also incorporates a programmed inclination to favor either younger or older individuals, adding variability to its responses across different scenarios.

The system evaluates groups of individuals, each varying in size from 1 to 10 members, with group composition randomly generated to simulate realistic conditions. The collective ethical value of a group is determined by the sum of its individual contributions, calculated through a combination of factors such as ethical weight, individual age, and a multiplier derived from the AI’s programmed preferences. This structured approach ensures that the AI's decisions are guided by a balanced consideration of group characteristics.

Operating without preconceived biases, the AI relies on a rule based algorithm to objectively evaluate the groups. The algorithm iteratively processes scenarios up to 10,000 in initial testing to determine which group to sacrifice, aiming to minimize ethical cost. Each iteration introduces unique combinations of group sizes, ages, and ethical weights, ensuring a diverse dataset for analysis.

To ensure the simulation adequately represents varied decision making contexts, it employs multiple AI instances, each independently processing its own set of scenarios. This diversity in AI behavior, combined with the stochastic nature of the parameters, ensures that no two simulations are identical. By blending structured rules with random variability, the simulation provides a robust framework for exploring the ethical capabilities of autonomous systems in complex moral dilemmas.

Preliminary Results

Through 10,000 iterations of the simulation, we examined the behavior of the AI in response to diverse ethical scenarios. The outcomes, represented in terms of scores and the choices made by the machine, were systematically analyzed to elucidate the influence of the identified variables on ethical decision making. To ensure statistically robust data, we subsequently conducted an extended analysis involving 1,170,000 iterations, further enhancing the reliability and validity of our findings.

Code Overview

The **C++ code** presented below provides the full implementation of the simulation's decision making process.

```
#include <iostream>
#include <vector>
#include <cstdlib>
#include <ctime>
/*
This code simulates the AI of an autonomous vehicle that must make an ethical decision based on an evaluation algorithm. The trolley problem is modeled through two groups of people, A and B, and the AI must decide which group to sacrifice in order to maximize its ethical score. The algorithm takes into account the ethical weight of the machine and variables such as the age of individuals in the groups. The goal is to explore the AI's behavior in different scenarios, considering its choices and the outcomes obtained. In order to achieve objective results, no particular ethical principles, such as human-like priorities (which favor saving younger individuals), have been imposed. Instead, the machine is allowed to analyze the opposite case, resulting in purely mathematical outcomes without any bias.
*/

// Define Macchina Struct
 struct Macchina {
	int punteggio;
	float PesoEtico;
	float MoltEta;
	float risultatoA = 0;
	float risultatoB = 0;
	bool Priori;
	std::string scelta;
 };

// Function to calculate the machine score for an individual in Group A
 float CalcolaRisultatoA(const Macchina& macchina, const std::vector<int>& gruppoA, bool priori) {
	float risultatoA = 0;
	for (size_t i = 0; i < gruppoA.size(); ++i) {
		// Calculation of the MoltEta value for Group A
		float moltEtaA = priori ? (5.0f - (gruppoA[i] / 100.0f) * 4.0f) : (1.0f + (gruppoA[i] /
		100.0f) * 4.0f);
		risultatoA += gruppoA[i] * macchina.PesoEtico * moltEtaA;
	}
	return risultatoA;
}

// Function to calculate the machine score for an individual in Group B
float CalcolaRisultatoB(const Macchina& macchina, const std::vector<int>& gruppoB, bool priori) {
	float risultatoB = 0;
	for (size_t i = 0; i < gruppoB.size(); ++i) {
	// Calculation of the MoltEta value for Group B
	float moltEtaB = priori ? (5.0f - (gruppoB[i] / 100.0f) * 4.0f) : (1.0f + (gruppoB[i] / 100.0f) *
	4.0f);
	risultatoB += gruppoB[i] * macchina.PesoEtico * moltEtaB;
	}
return risultatoB;
}
```

```
int main() {
	// Initialization of the random number generator
	std::srand(static_cast<unsigned>(std::time(0)));
	std::string uccisione;
	// Creation of a population of 10 machines
	const int numMacchine = 10;
	std::vector<Macchina> popolazione(numMacchine);

	for (int iterazione = 1; iterazione <= 10000; ++iterazione) {

	// Fill the population array with random values
		for (int i = 0; i < numMacchine; ++i) {
		popolazione[i].punteggio = 0;
		popolazione[i].PesoEtico = (static_cast<float>(std::rand() % 5) + 1); // Valore tra 0 e 10
		popolazione[i].MoltEta = 0; // Inizializzato a 0
		popolazione[i].Priori = (std::rand() % 2 == 0); // Scegli casualmente vero o falso
		}

	// Creation and filling of array A
	 const int grandezzaA = std::rand() % 10 + 1;
	std::vector<int> A(grandezzaA);
		for (int i = 0; i < grandezzaA; ++i) {
		A[i] = std::rand() % 76 + 5; // Valore tra 5 e 80
		}
	// Creation and filling of array A
	 const int grandezzaB = std::rand() % 10 + 1;
	std::vector<int> B(grandezzaB);
		for (int i = 0; i < grandezzaB; ++i) {
		B[i] = std::rand() % 76 + 5; // Valore tra 5 e 80
		}
	// Calculate the score for each machine individual and print the results

	 for (auto& macchina : popolazione) {
	macchina.risultatoA = CalcolaRisultatoA(macchina, A, macchina.Priori);
	macchina.risultatoB = CalcolaRisultatoB(macchina, B, macchina.Priori);

	// Compare the results and update the score and the choice
	if (macchina.risultatoA > macchina.risultatoB)
	{

	if(A.size() > B.size())
	{
	uccisione = "0 ";
	macchina.punteggio += A.size();
	}

	if(A.size() < B.size())
	{
	uccisione = "1 ";
	macchina.punteggio += B.size();
	}
```

```
macchina.scelta = "Uccide il gruppo B ";
}
else if (macchina.risultatoB > macchina.risultatoA)
{
if(A.size() < B.size()){
uccisione = "0 ";
macchina.punteggio += B.size();
}
if(A.size() > B.size()){
uccisione = "1 ";
macchina.punteggio += A.size();
}
macchina.scelta = "Uccide il gruppo A ";
}
else
{
macchina.scelta = "Nessuna scelta ";
}
if(A.size() == B.size()){
uccisione = " ";
macchina.punteggio += A.size();
}
// Print Results
std::cout << uccisione << std::endl;
}

}
return 0;
}
```

Data Following 1,170,000 Iterations:

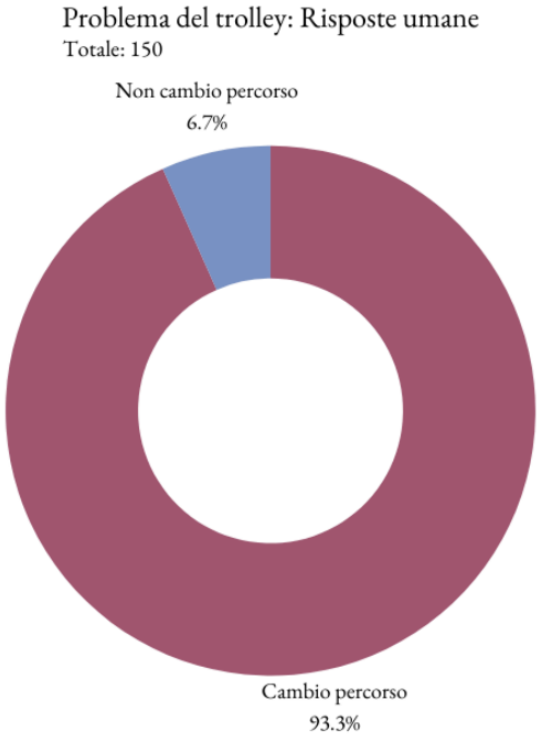
Iterazioni totali	Risulati: 0	% 0	Risultati: 1	% 1
90.000,00	82.548,00	91,72%	7.452,00	8,28%
90.000,00	82.998,00	92,22%	7.002,00	7,78%
90.000,00	82.665,00	91,85%	7.335,00	8,15%
90.000,00	82.926,00	92,14%	7.074,00	7,86%
90.000,00	82.683,00	91,87%	7.317,00	8,13%
90.000,00	82.620,00	91,80%	7.380,00	8,20%
90.000,00	82.809,00	92,01%	7.191,00	7,99%
90.000,00	82.755,00	91,95%	7.245,00	8,05%
90.000,00	82.674,00	91,86%	7.326,00	8,14%
90.000,00	82.809,00	92,01%	7.191,00	7,99%
90.000,00	82.782,00	91,98%	7.218,00	8,02%
90.000,00	82.476,00	91,64%	7.524,00	8,36%
90.000,00	82.332,00	91,48%	7.668,00	8,52%
1.170.000,00	1.075.077,00	91,89%	94.923,00	8,11%

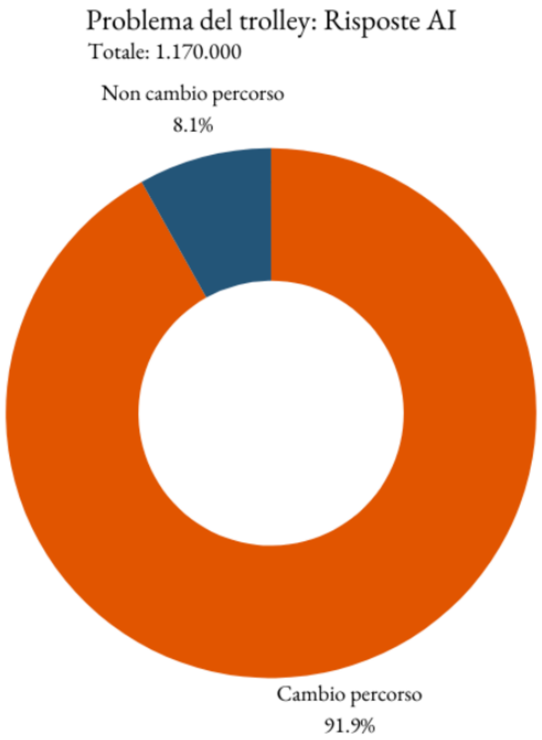
Dev st.	Dev st.
180,43	180,43
0,22%	2,47%

0 = Change the path
1 = Stay on the path

Conclusions

A crucial aspect of our study was the comparison between the ethical decisions made by the AI and the human preferences gathered through a preliminary questionnaire. Before implementing the simulator, we interviewed 150 individuals, asking them what choice they would make in similar situations, varying the number of people involved. Surprisingly, 93% of the participants indicated that they would choose to change the trolley's course when a larger number of individuals were at stake. Even more significant is that during our simulations, the machine opted to change the trolley's course at the same rate of 93%. This alignment between human preferences and the AI's decisions underscores the effectiveness of our simulator in capturing the ethical inclinations of individuals. The importance of this alignment between the simulation and human preferences lies in the fact that we have obtained a model that faithfully reflects the choice variables of people. This result suggests that our simulator is capable of successfully replicating human decision making dynamics in complex ethical contexts, offering valuable insights into how AI can reflect and respond to societal preferences in critical situations. In conclusion, our simulation not only provides a detailed framework of the ethical decisions made by the AI but also demonstrates its ability to adequately replicate the will and choice variables of individuals. This advancement is crucial for developing autonomous systems that align with societal preferences, thereby contributing to ethically responsible and socially accepted autonomous driving





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